

Topic A Part 2 Review [108 marks]

1. [Maximum mark: 20]

The police are considering changing from a relational database to either a network database or an object-oriented database.

(a) Identify **two** characteristics of a network database. [2]

(b) Discuss the advantages **and** disadvantages of an object-oriented database compared to a relational database. [5]

The European Union Agency for Law Enforcement Cooperation (Europol) may develop a data warehouse to hold data related to crimes.

(c) Outline **two** advantages of creating a data warehouse to store crime data. [4]

Predictive modelling is a technique that could be applied to the data warehouse.

(d) Outline **two** benefits of applying predictive modelling to crime data. [4]

Europol would like to apply link analysis to investigate fraud, uncover terrorist networks and investigate criminal activities.

(e) Explain how link analysis can be applied to crime datasets. [5]

2. [Maximum mark: 18]

The Bucharesti School website allows parents to login and select school transportation for their children. If they select the school bus, they will have to pay for this service at the end of the month.

- (a) Identify the steps that take place in a transaction when a parent attempts to pay for the school bus at the end of a month. [3]
- (b) Explain how the database management system (DBMS) prevents a record being updated by two parents simultaneously. [3]

The students' personal details must be securely stored in the database.

- (c) Identify **two** roles of the database administrator at Bucharesti School. [2]
- (d) Outline **two** ways that a database management system (DBMS) can be used to ensure the students' personal data remains secure. [4]
- (e) Explain how the developers of the Bucharesti School database can ensure that it has been designed ethically. [6]

3. [Maximum mark: 20]

Books are sold in physical bookshops and by online retailers. Each retailer maintains a unique database of books sold.

Online retailers can have a much greater range of books available. Books that are very rarely sold are not stocked and can be printed on demand.

Information from each retailer's database is loaded into a data warehouse where data analytics take place.

- (a) Outline **one** advantage of using a data warehouse. [2]
- (b) Explain why the data from the individual retailers is transformed before it is loaded into a data warehouse. [3]
- (c) Explain why data warehouses use timestamping. [3]

Predictive modelling uses decision trees and neural networks.

- (d) Distinguish between decision trees and neural network learning algorithms. [2]

Different data mining techniques are used to discover patterns.

- (e) Outline why deviation detection is used in data analytics. [4]

- (f) Compare cluster analysis and forecasting as techniques to understand and predict data in data mining. [6]

4. [Maximum mark: 15]

The restaurant owner has hired a database administrator to extend the database.

The database administrator has decided to delete the LinePrice field from the ORDERLINE table.

ORDERLINE

<u>OrderID</u>	<u>ItemID</u>	Quantity	UnitPrice	LinePrice
DM1007	F1234	1	16.99	16.99
DM1007	W2134	2	6.99	13.98

- (a) Explain why the decision to delete the LinePrice field has been made. [3]

The database administrator has created new tables to store employee data and create a work schedule.

When an employee works a shift in the restaurant, their hours and pay rate are recorded by the shift manager as a database transaction.

(b) Define the term *database transaction*. [1]

(c.i) State **one** way to validate the hours_worked field. [1]

(c.ii) Outline **one** way to verify the payrate field. [2]

The shift manager sometimes changes the employees' work hours in the database during their shift. Employees can access the work hours and confirm their availability.

(d) Explain how data integrity can be maintained if an employee tries to access their work hours while the shift manager is editing them. [4]

The database administrator has set up a procedure for database recovery.

One method of recovering a database is to use a transaction log.

(e) Describe **one** alternative method that could be used to recover a database after a crash has occurred. [4]

5. [Maximum mark: 15]

A European pet passport allows pets to travel between countries in the European Union. All pets have a microchip inserted that gives them a unique alphanumeric transponder code.

The data contained in a pet passport is shown in **Figure 1**. All of the data is stored in a database.

Figure 1: Two pages of a pet passport

DESCRIPTION OF ANIMAL	ISSUING OF PASSPORT
PICTURE OF THE ANIMAL (optional)	Name of the authorized veterinarian: Helen Bright
Name: Benji	Address: Lowden Bridge House, Lowther Street, Eskdale
Species: Dog	Postcode: CA37 8DX
Breed: Labrador	Country: United Kingdom
Sex: Male	Telephone number: 01632 960739
Date of birth: 01/01/2017	Email address: jkr@gov.example.com
Alphanumeric transponder code: 985 AAA 111 AAA 111	Date of issue: 26/01/2019 

[Source: Department of Agriculture, Environment and Rural Affairs. EU Pet Travel Scheme - How to Complete Pet Passports. <https://www.daera-ni.gov.uk/publications/eu-pet-travel-scheme-how-complete-pet-passports>. Contains public sector information licensed under the Open Government Licence v3.0. Source adapted <https://www.daerani.gov.uk/crown-copyright>.]

A pet owner may have more than one pet, and each pet has their own passport. Pet passports are issued by registered veterinary surgeons (vets).

- (a) Construct the entity-relationship diagram (ERD) that shows the relationship between pet owners, their pets, and the pet passport.

[2]

The data stored in the VET table can be represented in the following notation:

VET(VetID, FirstName, Surname, Address, Postcode, City, Country, Phone, Email)

The Phone field will be used as a secondary key.

(b) Define the term *secondary key*.

[1]

Each year, pets require vaccinations to prevent them from contracting rabies. Information about each vaccination can be seen in the pet passport (see **Figure 2**).

Figure 2: Vaccination information page in a pet passport

VACCINATION AGAINST RABIES			
MANUFACTURER AND NAME OF VACCINE	Place vaccine sticker or handwrite: Merck Animal Health, Nobivac, 2021		
BATCH NUMBER			
1. VACCINATION DATE	1. 05/01/2019	2. 26/01/2019	3. 05/01/2021
2. VALID FROM			
3. VALID UNTIL			
AUTHORIZED VETERINARIAN	Name, address, telephone number, signature and SP number		

[Source: Department of Agriculture, Environment and Rural Affairs. EU Pet Travel Scheme - How to Complete Pet Passports. <https://www.daera-ni.gov.uk/publications/eu-pet-travel-scheme-how-complete-pet-passports>. Contains public sector information licensed under the Open Government Licence v3.0. Source adapted <https://www.daerani.gov.uk/crown-copyright>.]

The information in **Figure 1** and **Figure 2** needs to be normalized before being entered into the database.

The VET table is given and will need to be incorporated into the final structure.

(c) Construct the database in 3rd Normal Form (3NF) for all of the entities shown in **Figure 1** and **Figure 2**.

You should use the same notation as that used in the representation of the VET table.

[8]

Access to this database must comply with data protection legislation.

- (d) Describe the principles of data protection legislation that would apply to a pet owner's data.

[4]

6. [Maximum mark: 20]

An emergency management information system (EMIS) uses a spatial database to map all emergency calls via their coordinates. Every emergency call generates data that includes spatial data.

- (a.i) Outline what is meant by the term *spatial data*.

[2]

- (a.ii) State **two** types of geometric object that can be represented by a spatial database.

[2]

Spatial data on a map are always linked to a row in a table.

- (b) State **one** type of relationship that can be used to represent these links.

[1]

The map in **Figure 3** shows the locations of crimes reported in Los Angeles, USA, in 2019.

The larger the dot, the more crimes that have been reported in that area.

Figure 3: Map showing the locations of crimes reported in Los Angeles, USA, in 2019



[Source: Wikipedia, n.d. *California locations by crime rate*. [online] Available at: https://en.wikipedia.org/wiki/California_locations_by_crime_rate [Accessed 11 January 2021] and Wikimedia Commons, 2008. *Locator map of the Harbor Freeway in Los Angeles County, Southern California*. [online] Available at: https://commons.wikimedia.org/wiki/File:Harbor_Freeway.svg [Accessed 11 January 2021]. Source adapted.]

Data mining techniques can be applied to the spatial data on the map to provide additional data analysis.

- (c) Explain how **two** data mining techniques could be applied to provide a different view of the data displayed on this map. [6]

Visual data analysis tools can be used to make the data on maps appear differently. For example, two different political parties may want the map data to show their different perspectives.

- (d) Discuss the ethics of using visual data analysis tools. [5]
- (e) Describe how deviation detection can be used to predict potential incidents, such as fires. [4]